ModelIt!

Teacher's Guide

The Invisible Plant Factory: Photosynthesis Deeper

How Plants Build Themselves from Thin Air and Sunlight

Quick Reference

Grade Level	Grade 06L2
NGSS Standard	MS-LS1-6, MS-LS1-7
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-LS1-6, MS-LS1-7

Construct a scientific explanation based on evidence for the role of photosynthesis in the cycling of matter and flow of energy into and out of organisms. Develop a model to describe how food is rearranged through chemical reactions forming new molecules that support growth and/or release energy as this matter moves through an organism.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model with three external inputs showing how light, CO_2, and water interact to produce glucose, oxygen, and plant growth.
Disciplinary Core Idea	**LS1.C Organization for Matter and Energy Flow in Organisms** — Plants use the energy from light to make sugars from carbon dioxide and water through photosynthesis. This produces food that can be used immediately or stored for later use.
Crosscutting Concept	**Energy and Matter: Flows, Cycles, and Conservation** — Students trace the flow of matter (carbon from CO_2 becomes glucose becomes plant mass) and energy (light energy becomes chemical energy in glucose) through photosynthesis.

CAST Testing Alignment

MULTIPLE CHOICE:

A scientist grows two identical plants in sealed containers. Plant A receives normal light, CO_2, and water. Plant B receives normal light and water but the CO_2 has been removed from the air. After 4 weeks, which result would you most likely observe?

CONSTRUCTED RESPONSE:

A student claims that plants get their food from the soil through their roots. Using your photosynthesis model and the famous Jan van Helmont willow tree experiment, explain why this claim is mostly incorrect. Where does the mass of a plant actually come from?

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Photosynthesis is the foundation of almost all life on Earth. This single chemical process captures solar energy, removes CO_2 from the atmosphere, produces the oxygen we breathe, and creates the food that feeds nearly every food web. Understanding it deeply reveals the elegant chemistry that powers our planet.

The Photosynthesis Equation

$6CO_2 + 6H_2O$ + light energy yields $C_6H_{12}O_6 + 6O_2$. Six molecules of carbon dioxide combine with six molecules of water, using light energy, to produce one molecule of glucose and six molecules of oxygen. Every atom on the left appears on the right. Mass is conserved. Energy is transformed from light to chemical energy stored in glucose bonds.

Where Plant Mass Really Comes From

In 1648, Jan van Helmont grew a willow tree for 5 years. The tree gained 164 pounds, but the soil lost only 2 ounces. He concluded the mass came from water, but we now know most plant mass comes from CO_2 in the air. Carbon atoms from atmospheric CO_2 are fixed into glucose, then assembled into cellulose, lignin, and other structural molecules. A tree is literally made of air.

Three Inputs, One Limiting Factor

Photosynthesis requires light, CO_2, and water simultaneously. The rate is limited by whichever input is in shortest supply. In a bright greenhouse with plenty of water, CO_2 concentration is often the limiting factor. In a desert with abundant light and CO_2, water is the limiting factor. Commercial greenhouses supplement CO_2 to levels 2-3 times atmospheric concentration to overcome this limitation.

Photosynthesis and Cellular Respiration Are Opposites

Photosynthesis builds glucose from CO_2 and H_2O using light energy. Cellular respiration breaks glucose down into CO_2 and H_2O, releasing the stored energy as ATP. They are essentially the same reaction running in opposite directions. Plants perform BOTH processes: photosynthesis during the day and cellular respiration 24 hours a day.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify light intensity, CO_2 concentration, and water availability as the three external inputs, and glucose production, oxygen released, and plant biomass growth as the internal responses.

E **Establish Relationships**

Students determine that all three inputs positively drive glucose production, glucose production positively drives both oxygen release and plant biomass growth, and the overall rate is limited by whichever input is lowest.

V **Visualize & Model**

Students build a model with three independent inputs converging on one central process (photosynthesis), demonstrating that ALL inputs are required and the limiting factor controls the rate.

E **Evaluate Outcomes**

Students run full sun, darkness, and drought scenarios to observe how removing different inputs affects the system differently, and discover the concept of limiting factors.

R **Revise & Extend**

Students add temperature or stomata openness to explore the tradeoff between CO_2 uptake and water loss that plants face in hot, dry environments.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"This leaf is a solar-powered factory that builds itself from thin air. Literally. And it has been doing this for 3 billion years."*

Watch For

Hold up a leaf. Tell students it is a factory that takes invisible gas and turns it into solid wood. Challenge them to figure out how.

Slide 2: Learning Objectives

3 min

Say: *"Today we are going to discover that trees are made of AIR, and prove it with a model."*

Watch For

Have students read objectives. Pre-teach 'glucose,' 'carbon fixation,' and 'limiting factor.'

Slide 3: Big Question

3 min

Say: *"A redwood weighs 2 million pounds. It started as a seed smaller than your fingernail. Where did all that mass come from?"*

Watch For

Poll the class: soil, water, air, or sunlight? Most will say soil. They will be surprised.

Slide 4: LEVER Framework

2 min

Say: *"Our model has THREE external inputs. This is the most complex model we have built. All three must be present for the system to work."*

Watch For

Preview the three-input structure. Tease that removing ANY one input crashes everything.

Slide 5: Activity 1: Components

8 min

Say: *"What does a plant need from outside? What does it produce inside? Notice we have SIX components."*

Watch For

Guide sorting. Discuss why all three inputs are external (supplied by the environment, not made by the plant).

Slide 6: Activity 2: Connections

8 min

Say: *"Light, CO_2, and water all converge on one process. Glucose is the product, and from glucose comes oxygen and growth."*

Watch For

Write the photosynthesis equation. Match each part to a model component. Count atoms on both sides.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's turn off the lights and watch the factory shut down. Then let's create a drought. Which kills the plant faster?"*

Watch For

Students predict before running. Compare darkness (immediate stop) to drought (gradual decline). Discuss limiting factors.

Slide 8: Discoveries

5 min

Say: *"A tree is made of air. The carbon in CO_2 becomes the carbon in wood. You are breathing tree-building material right now."*

Watch For

Share van Helmont's willow experiment. Connect CO_2 to plant carbon to cellulose to wood. Mind-blowing for students.

Slide 9: STEM Challenge

12 min

Say: *"Design a factory that makes food from air, water, and light. It is called a greenhouse. Make it the best one possible."*

Watch For

Distribute materials. Groups design growth chambers optimizing all three inputs.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Light Intensity, CO2 Concentration, Water Availability
- **INTERNAL:** Glucose Production, Oxygen Released, Plant Biomass Growth

Relationships

- **Light Intensity to Glucose Production:** POSITIVE (+) — More light energy powers more photosynthesis reactions, producing more glucose. Without light, the light-dependent reactions cannot split water or generate the energy carriers needed for carbon fixation.
- **CO2 Concentration to Glucose Production:** POSITIVE (+) — More CO2 provides more carbon atoms for fixation into glucose. Carbon from CO2 becomes the carbon backbone of every glucose molecule. Higher CO2 means faster fixation.
- **Water Availability to Glucose Production:** POSITIVE (+) — Water provides hydrogen atoms and electrons for the light reactions. Without water, the plant cannot split H2O molecules, and the entire process stops.
- **Glucose Production to Oxygen Released:** POSITIVE (+) — Oxygen is a byproduct of splitting water molecules during photosynthesis. More photosynthesis means more water split and more O2 released.
- **Glucose Production to Plant Biomass Growth:** POSITIVE (+) — Glucose is used to build cellulose, lignin, proteins, and all other structural molecules. More glucose production means more building material for growth.

Key Discoveries

- Photosynthesis requires ALL THREE inputs: light, CO2, and water. Removing any one of them stops the entire process
- Glucose production is limited by whichever input is in shortest supply (the limiting factor), not by the most abundant one
- Most of a plant's mass comes from carbon dioxide gas in the air, NOT from soil. Trees are made of air!
- Oxygen is a BYPRODUCT, not the purpose of photosynthesis. Plants make oxygen because they must split water molecules
- The photosynthesis equation (6CO2 + 6H2O + light -> C6H12O6 + 6O2) shows matter conservation: same atoms in and out

Multiple Choice Post-Assessment Answer Key

1. A giant redwood tree weighs over 2 million pounds. Where did most of that mass come from?

Correct: C) Carbon dioxide gas from the atmosphere, converted to glucose and then to wood through photosynthesis

Correct! Through photosynthesis, plants take carbon atoms from CO2 in the air and fix them into glucose molecules. That glucose is then assembled into cellulose, lignin, and other structural molecules that make up wood. A tree is literally built from air. Soil minerals make up less than 5% of tree mass.

2. In the photosynthesis model, what happens when light is abundant and water is plentiful but CO2 is very low?

Correct: B) Glucose production is severely limited because CO2 is the limiting factor, regardless of how much light and water are available

Correct! This demonstrates the limiting factor concept. Photosynthesis requires ALL three inputs simultaneously. The rate is controlled by whichever input is in shortest supply. With low CO2, there are not enough carbon atoms to fix into glucose, so production is limited regardless of how much light and water are available.

3. Why is oxygen described as a 'byproduct' of photosynthesis rather than its main purpose?

Correct: C) Oxygen is released when water molecules are split to provide hydrogen and electrons for glucose production. The plant needs the hydrogen, not the oxygen.

Correct! During the light reactions of photosynthesis, water molecules (H2O) are split to harvest hydrogen atoms and electrons needed to build glucose. The oxygen atoms left over are released as O2 gas. The plant does not make oxygen on purpose. It is a leftover from splitting water for its useful parts.

4. How are photosynthesis and cellular respiration related?

Correct: B) They are essentially opposite reactions: photosynthesis builds glucose from CO_2 and water using light energy, while cellular respiration breaks glucose down into CO_2 and water to release energy

Correct! Photosynthesis and cellular respiration are reverse processes. Photosynthesis: $CO_2 + H_2O$ + light energy yields glucose + O_2. Cellular respiration: glucose + O_2 yields $CO_2 + H_2O$ + ATP energy. The products of one are the reactants of the other. Plants actually perform BOTH processes.

5. A scientist grows two identical plants in sealed containers. Plant A has normal light, CO_2, and water. Plant B has normal light and water but the CO_2 has been removed. After four weeks, what would the model predict?

Correct: C) Plant B barely grows because without CO_2, it cannot perform photosynthesis or produce glucose for building new tissue

Correct! CO_2 provides the carbon atoms that plants fix into glucose and then into all structural molecules. Without CO_2, photosynthesis cannot produce glucose. Without glucose, the plant has no building material for growth and no energy source. Plant B would barely grow and would slowly consume its stored reserves.

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Plants get their food from the soil"	Plants get only water and minerals from the soil. Their actual food (glucose) is made through photosynthesis from CO_2 in the air and water, using light energy. Soil nutrients make up less than 5% of plant mass. Van Helmont proved this in 1648.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Plants breathe in CO_2 and breathe out O_2 (like reverse animals)"	Plants perform BOTH photosynthesis and cellular respiration. During the day, photosynthesis is faster, so they appear to absorb CO_2 and release O_2. At night (no light), they only do cellular respiration, absorbing O_2 and releasing CO_2. Plants are not reverse animals; they do both processes.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Sunlight is the plant's food"	Sunlight is ENERGY, not food. Light provides the energy to drive the chemical reaction that makes food (glucose). It is like saying electricity is your food because it powers your stove. The food is the glucose molecule, and light is the energy that assembles it from CO_2 and water.